FSD

Unit Test - 1

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CSE - A

1. Explain the essential technologies in full stack development with an architecture.
2. Compare the ORM tools and hibernate

Answers

1) Full stack Development involves building both the frontend and backend of a web application.

Essential technologies:

1. Frontend:

Html, CSS, JavaScript, — For structure, styling and interactivity.

* Frameworks : React, Angular, vue.js

2. Backend:

* Language / frameworks - Node.js, Express.js, Django, Spring Boot.
* handles business logic and API creation.

3. Database:

* SQL or NoSQL

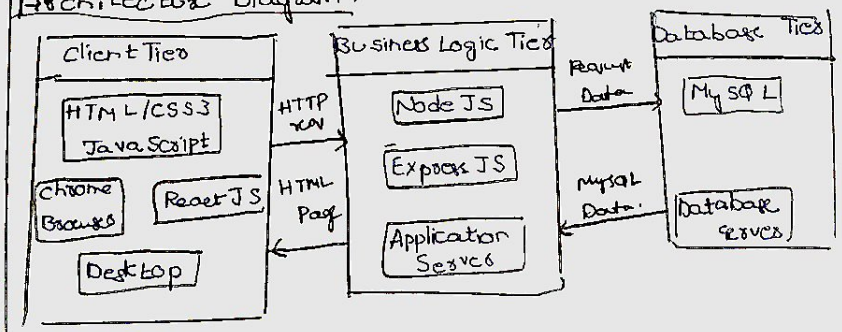
4. Version Control:

Git, Github - for tracking code changes

5. Deployment tools:

* Dockers, AWS, Heroku, vercel - for hosting and deployment.

Architecture Diagram:



2) Compare the ORM tools and hibernate.

Aspect	ORM (General)	Hibernate (Specific ORM)
Defination	ORM (Object Relational mapping) is a concept or technique that maps objects in programming languages to database tables	Hibernate is a specific implementation of ORM for java.
Scope	General approach used in many languages	Only works with java based applications.
Purpose	Simplifies database operations using object oriented code	Provides java developers a powerful ORM framework.
Examples	Hibernate (Java), Django ORM (Python), Entity framework (C#)	Hibernate is one of the most popular ORM tools in Java
Features	Depends on the specific ORM tool used	Supports HQL, Lazy loading.